

Blockchain based key agreement

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- Traditional centralized server architecture for storing health related patient records problematic
- Management of health data critical
- The need to protect credentials from unauthorized parties

The Problem

- Centralized server single point of failure
- Hard to share data between hospitals
- Risk of cyberattacks

The Proposed Solution

- Blockchain based key authentication scheme
- All information is stored in the blockchain ledger immutable and secure
- Authentication for IoT devices, patients and doctors
- Session key during communication for preventing session hijacking

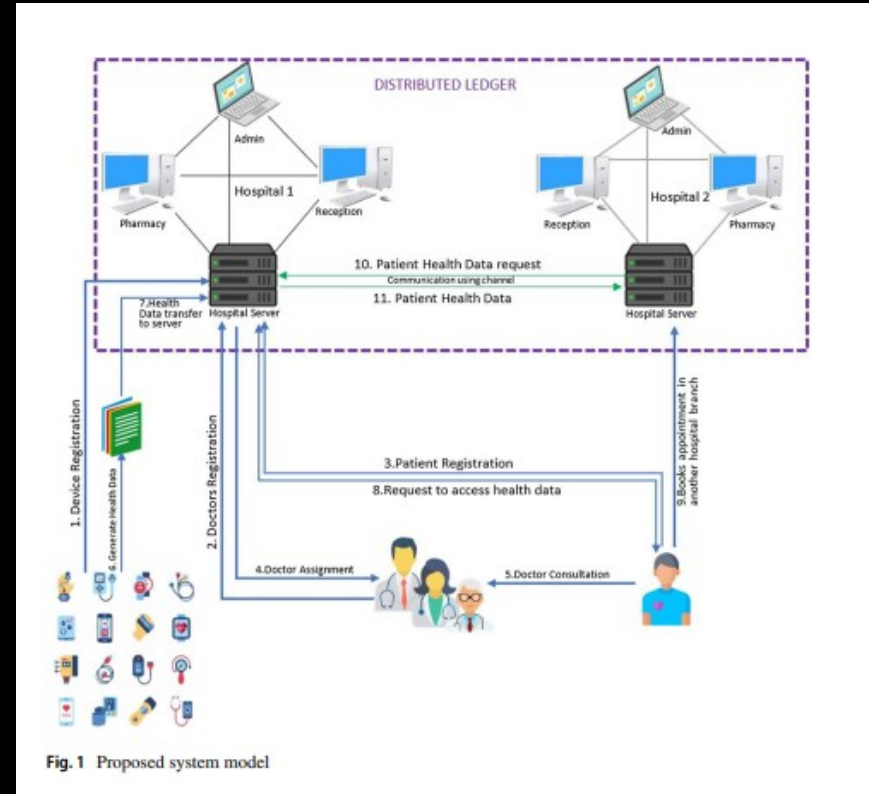


Fig. 1 Proposed system model

System Architecture

- Peer nodes :
Hospital Departments/Edge devices
(ledger and smart contracts)
- Orderer nodes:
Hospital Servers
(Order transactions and form blocks)
- Client nodes:
Applications
(submit and query transactions)

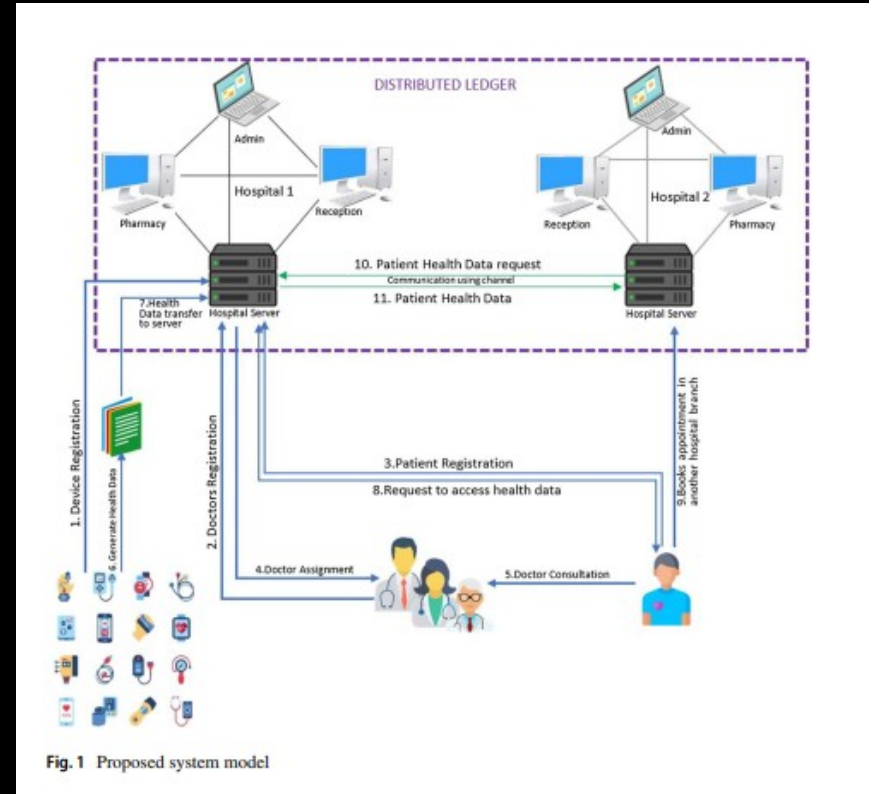


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Authentication Process

- Registration
- 3 factor authentication username password and biometrics → authentication token
- Session key for secure communication
- (Credentials stored in blockchain)

Security Features

The system protects from

- Replay attacks
- Impersonation
- Man in the middle attacks

Tools used for testing

- ROR Model
- Scyther

Technologies Used

- Blockchain: Hyperledger Fabric, Raft consensus , smart contracts
- Cryptography: ECC
- Infrastructure: Docker, Kubernetes, CouchDB
- Authentication: 3-factor authentication

- Blockchain-based Dynamic Group key agreement
- Non-centralized authentication ,privacy
- Distributed computing and communication
- Dynamic BD-GKA protocol forward and backward secrecy security

Core Idea

- BD-GKA protocol with Diffie-Hellman key exchange
- Anonymous authentication
- Traceability
- Dynamic membership (no group restarts)
- Self-certified group keys

Components

- Key Generation Center
- Nodes
- Blockchain
- Group Key

Key Generation Center

- Initializes system parameters
- Verifies identities
- Distributes private/public keys

Protocol Algorythm

- Initialization
- Group Key Computation
- Join Phase
- Leave Phase

Initialization

- Set up parameters (bilinear groups ,hash functions , master keys)
- Each Node gets a public and private key based on id

Group Key Computation

- Each node generates a random number and computes a signature using bilinear maps
- Partial keys and hashes go to the blockchain
- The last node incorporates all values to make the final group key
- The last node broadcasts the key
- Every Node verifies it

Node Join

- New node broadcast its ip and public key
- An updated transaction chain recorded by the blockchain
- New group key with new random values for forward secrecy

Node Leave

- Update gets triggered
- Remaining nodes recompute a new key

- 5G introduces URLLC, mMTC and eMBB → higher security demands
- Traditional 5G-Authentication and Key agreement assumes secure channel between Home and Serving Network
- That assumption is unrealistic and the channel prone to attacks

Problem definition

- Secure HN-SN link
- Privacy leak via sequence numbers
- Replay and impersonation attacks
- Malicious Sns possible during roaming

Proposed Solution

- Blockchain-based 5G-AKA Protocol
- Each home network creates a smart contract on blockchain
- Serving network communicates with it for authentication
- Removes direct HN-SN channel
- Uses encrypted transaction metadata to carry messages securely

Architecture

User Equipment (mobile or IoT)
Serving Network (visited network)
Home network
Smart Contract on blockchain

Security Features

- No need for secure-channel dependency
- Preserves privacy: keys encrypted end to end
- Prevents replay, impersonation and Dos attacks

Implementation

- Ethereum Smart Contracts

Advantages

- Decentralized
- User anonymity
- Eliminates secure channel dependency
- Enables cryptocurrency payments

Trade-offs

- Latency due to blockchain transactions
- Ethereum gas cost
- Slightly higher setup complexity

Conclusion

- Blockchain can replace secure channels in 5G-AKA
- Provides verifiable, privacy-preserving roaming authentication
- Future: use private/consortium blockchains for reduced delay

- IoT networks contain millions of low-power, resource-limited devices.
- Traditional authentication (e.g., PKI) = too heavy and centralized.
- Many schemes fail against replay or impersonation attacks.
- Need: a lightweight, decentralized, and scalable authentication mechanism.

Problem definition

- Centralized authentication → single point of failure.
- Limited computation & storage on IoT devices.
- Weak key management in distributed systems.
- High latency and poor scalability in PKI-based solutions.

Solution-Main idea

- Three phases: Registration → Authentication → Key Agreement.
- Blockchain acts as public verification ledger — no need for central authority.
- ECC cryptography → efficient and secure identity & key exchange.
- Smart contracts handle device registration, validation, and log events.

Security

- Based on Elliptic Curve Discrete Logarithm Problem (ECDLP).
- Mutual authentication
- Anonymity and unlinkability
- Forward and backward secrecy
- Replay and impersonation resistance
- Verified with AVISPA or ProVerif simulation tools.
- No central authority = no single point of failure.

Blockchain

- Stores hashed identities and session proofs (not plaintext keys).
- Smart contract verifies device IDs and timestamps.
- Prevents data tampering and replay attacks.
- Enables auditable, verifiable trust without central servers.

Technologies

- ECC
- Hypeledger Fabric
- AI-driven anomaly detection

- IoT devices have limited CPU, memory, and power.
- Traditional authentication = heavy, centralized, prone to attack.
- IoT needs lightweight, scalable, and decentralized security.
- Blockchain can add trust and transparency without central authority.

Problem Definition

- Centralized PKI and key management → single point of failure.
- High communication and storage overhead for IoT nodes.
- Existing ECC schemes vulnerable to impersonation and replay attacks.

Solution

- Blockchain-Based Lightweight Authentication and Key Agreement (BLAKA):
- Combines blockchain ledger + ECC cryptography.
- Provides mutual authentication and session key agreement.
- Registration, authentication, and key updates recorded on blockchain.
- Smart contracts used for verification and event logging.

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System Architecture

- IoT devices
- Gateway → verifies and authenticates IoT
- Blockchain Network
- Cloud/Authority → registers gateways and manages contract deployment

Protocol

- Registration
- Authentication
- Key Agreement

Security

- Based on ECC security
- Resists replay, impersonation, mitm and session key disclosure attacks
- Ensures anonymity, mutual authentication, forward secrecy

- IoT networks are highly heterogeneous and resource-constrained.
- Centralized models (cloud-based) face issues of:
 - High latency & bandwidth demand
 - Privacy and single point of failure
- Need for decentralized, scalable, and secure IoT communication.

Solution

- Integrates Hyperledger Fabric (HLF) for secure IoT data exchange.
- Uses clustering (GA + SA algorithms) for scalability & energy efficiency.
- Combines local (private) and global (public) blockchains.

Architecture

- Layer 1: IoT Devices (nodes, sensors, actuators)
- Layer 2: Cluster Heads (CHs) – local Hyperledger blockchain
- Layer 3: Base Stations (BS) – global blockchain (Ethereum-like)

Technologies

- Hyperledger Fabric
- ECC
- Genetic Algorithm + Simulated Annealing for clustering
- Docker + CouchDB

- Smart grids involve millions of connected devices (smart meters, sensors).
- Existing models are centralized → single point of failure.
- Cloud-only systems cause high latency and slow response.
- Need for distributed, low-latency, and secure solutions.

Solution

- Three-layer system:
- Smart Meters (SM): measure and send data.
- Fog Nodes (FN): near-edge controllers, reduce latency.
- Cloud Servers (CS): manage blockchain ledger and coordination.
- Blockchain framework: Hyperledger Fabric

Solution

- Blockchain-Assisted Authentication and Key Agreement
- Three-party mutual authentication: SM $\leftrightarrow$ FN $\leftrightarrow$ CS
- Uses ECC (Elliptic Curve Cryptography) for lightweight security.
- TA (Trusted Authority) registers all nodes and initializes blockchain.
- Each node has pseudo-identity to protect privacy.

- X25519 (for Elliptic Curve Diffie–Hellman key exchange)
- Ed25519 (for digital signatures)
- Blockchain + Smart Contracts (for decentralized public key management)

Function

- Each IoT device generates two ECC key pairs (X25519 for exchange, Ed25519 for signing).
- Its public keys and unique ID are recorded immutably on the blockchain via a smart contract.
- Devices retrieve each other's verified public keys directly from blockchain.
- Use X25519 to derive a shared secret key for session encryption.
- Timestamps + HMACs ensure freshness and message integrity.
- Devices can securely update their public keys on blockchain with digitally signed requests, maintaining forward secrecy.

Security

Resistant to:

- Replay
- Impersonation
- Man-in-the-middle
- Key compromise attacks
- Supports forward secrecy and mutual authentication.

Technologies

- ECC
- Diffie-Hellman
- Ed25519
- Euthereum and Polygon

SUMMARY-CONCLUSIONS

- Proposes a blockchain–ECC–biometric protocol for secure medical data exchange between hospitals, IoT devices, and patients.
- Uses Hyperledger Fabric and three-factor authentication (ID, password, biometric).
- Ensures mutual authentication, forward secrecy, and privacy.
- Validated using ROR model and Scyther; highly efficient for healthcare IoT.

- Introduces a group key agreement scheme using bilinear pairing and blockchain for mobile ad hoc networks.
- Enables anonymous yet traceable group communication.
- Provides dynamic membership (nodes can join/leave securely).
- Achieves low computation cost and strong resistance to impersonation and replay attacks.

- Replaces the secure HN–SN channel in 5G-AKA with Ethereum smart contracts.
- Enables trustless roaming authentication and immutable verification among network operators.
- Verified with ProVerif, ensuring privacy, replay, and DoS resistance.
- Eliminates centralized trust but introduces slight latency due to blockchain transactions.

- Designs a lightweight ECC-based authentication scheme with blockchain as a trust anchor.
- Performs device registration, mutual authentication, and session key generation via smart contracts.
- Provides anonymity, forward secrecy, and decentralized verification.
- Reduces computation and communication cost while improving scalability.

- A lightweight mutual authentication protocol combining ECC and blockchain ledger.
- Removes reliance on centralized PKI while maintaining strong security.
- Proven secure against impersonation, replay, and man-in-the-middle attacks using AVISPA.
- Ideal for resource-constrained IoT devices due to low energy use and computation overhead.

- Implements a multi-layer blockchain model using Hyperledger Fabric for IoT.
- Divides the network into local (private) and global (public) blockchains.
- Integrates clustering (Genetic Algorithm + Simulated Annealing) for scalability and low latency.
- Achieves better throughput and lower latency than Ethereum; suitable for large-scale IoT networks.

- Develops a fog–cloud hybrid blockchain framework for smart grids.
- Uses Hyperledger Fabric and ECC for secure multi-party authentication.
- Achieves distributed, low-latency authentication with mutual trust among smart meters, fog nodes, and cloud.
- Proven secure under ROR model; experimentally efficient using Docker-based Hyperledger setup.

- BEdX25519 provides a lightweight, secure, decentralized authentication and key exchange mechanism for IoT D2D networks — combining blockchain transparency with ECC efficiency.